Does Female Leadership Relate to Productivity–Wage Gaps in Colombia?

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Abstract

In labor markets with frictions such as employer concentration, search costs, and limited worker mobility, firms can exercise monopsony power by setting wages below workers' marginal products, thereby generating productivity-wage gaps. The extent of these gaps depends not only on market structure but also on firm managerial structure, which shapes wage-setting and employment policies. A growing literature suggests that female leaders place greater weight on worker welfare and are less likely to engage in anticompetitive behavior. This paper examines whether these tendencies translate into reduced monopsony exploitation, specifically whether female leadership (measured as the share of female executives) attenuates the productivity-wage gap. Using firm-level panel data from the Colombian Annual Manufacturing Survey 2000-2019, I estimate firm-specific productivity-wage gaps following Yeh et al. (2022) and relate them to the female share of firm executives in a linear fixed-effects framework controlling for time-varying firm characteristics, geography, and industry-level factors. I find that increases in female leadership share are associated with reductions in the productivity-wage gap. These findings carry implications for both equity and efficiency: since wages below marginal product depress employment and distort resource allocation, expanding female leadership in emerging markets may therefore yield meaningful labor market gains.

JEL Classification: D43, D63, J16,J42,J7, M12, M52

Keywords: female leadership, labor market power, productivity-wage gap, labor monopsony, Emerging markets, Colombia

1 Introduction

Across the world, workers are paid substantially less than the value they produce. In Colombia, Amodio and de Roux (2024) estimates that workers produce approximately 40 percent more than they earn, a gap that represents not only foregone income for millions of workers, but also a significant source of allocative inefficiency for the broader economy. Understanding what can be done to reduce it is a central question in labor economics.

Standard microeconomic theory predicts that in competitive labor markets, wages equal workers' marginal revenue product. Yet a large and growing body of evidence shows that firms are not wage takers: labor markets are characterized by frictions — including search costs, limited job mobility, and employer concentration — that grant firms wage-setting power[1]. When firms exploit this power, they set wages below workers' marginal productivity, generating a productivity-wage gap commonly referred to as a labor markdown. The consequences extend beyond wages: firms with monopsony power also hire fewer workers than is socially optimal, depressing employment and distorting resource allocation across the economy.

Although monopsony power arises from structural features of labor markets, firms differ substantially in how aggressively they exploit it. This variation suggests that firm granularity and strategic wage setting shape the productivity-wage gap Berger et al. (2022). One such characteristic is the gender composition of firm leadership. A growing literature documents that female leaders tend to place greater weight on worker welfare and employment stability (Matsa & Miller, 2013; Tunyi et al., 2023), interpret productivity signals differently (Chen et al., 2019; Flabbi et al., 2017), and influence corporate decision-making in ways that may reduce productivity-wage distortions (Flabbi et al., 2019; Levi et al., 2014). At the same time, labor market power has been shown to affect women more strongly than men (Sharma, 2023), raising the possibility that female leadership may also attenuate gender-specific productivity-wage distortions within firms. Together, these strands of evidence motivate the central question of this paper: does greater female leadership reduce the extent to which firms exploit monopsony power?

To examine this question, I use firm-level panel data from the Colombian Annual Manufacturing Survey between 2000–2019. Colombia provides a particularly informative setting for this analysis. Its manufacturing sector exhibits substantial variation in both female executive representation and labor market concentration, and labor market frictions are

[1] A growing literature shows that labor markets are often characterized by frictions. See for instance on search costs (Card, 2022; Azar et al., 2019; Lentz and Mortensen, 2010; Moscarini and Postel-Vinay, 2010; Manning, 2021), on limited job mobility (Caldwell and Danieli, 2024; Schubert et al., 2025; Starr et al., 2021; McAdams, 2019), and on employer concentration (Azar et al., 2017; Benmelech et al., 2018; Rinz, 2022).

more pronounced than in high-income economies, making deviations from competitive wage-setting both more likely and more consequentialAlviarez et al. (2025). I estimate firm-specific productivity-wage gaps following the production-function methodology of Yeh et al. (2022) and then relate these estimates to the female share of firm executive positions. I use a linear fixed-effects framework that controls for time-varying firm characteristics and industry-level factors, isolating the within-firm relationship between management composition and wage-setting behavior.

The results reveal a negative association between female leadership share and the productivity-wage gap, consistent with less aggressive exploitation of monopsony power. This finding holds across a range of specifications and is not driven by differences in firm size, productivity, or industry composition. These results contribute to two distinct literatures. First, they add to the growing body of work on the determinants of labor market power by showing that management structure shapes the productivity-wage gap. Second, they extend the literature on female leadership and firm behavior by connecting gender composition in executive positions to a concrete labor market outcome with direct welfare implications. These findings carry important policy implications, including bias-free hiring, mentorship, flexible work, and pay equity.

2 Literature Review

Table 1: Summary of Related Literature

Paper	Topic	Data	Main Finding
De Loecker, Eeckhout, & Unger (2020)	Labor market concentration	U.S. manufacturing	Production approach for a robust estimation of elasticities. Findings suggest markups rise not just to cover higher fixed costs, but also due to firms' market power.
Yeh, Macaluso & Hershbein (2022)	Labor markdown estimation	U.S. manufacturing	Most manufacturing plants operate in a monopsonistic environment. Average markdown 1.53 workers earning only 65 cents on the marginal dollar generated.
Amodio & de Roux (2024)	Colombia Labor market power	Colombian manufacturing plants (EAM)	Estimate a firm-level labor supply elasticity of around 2.5, implying that workers produce about 40% more than their wage level
Inter-American Development Bank (2025)	Latin-America and the Caribbean Market Power	Firm-Data LAC (For Colombia EAM)	On average, workers in Latin America and the Caribbean take home in the form of wages 50 percent of what they contribute to their employers.
Matsa & Miller (2013)	Female leadership	Nordic companies (Norway's 2006 quota)	Female leaders tend to emphasize employment stability and worker retention.
Flabbi et al. (2019)	Female executives	Italy employer-employee panel data	Female leadership influences wage-setting and reduce gender gaps within firms.
Levi, Li & Zhang (2014)	Female executives	Aquisition bids by S&P 1500 companies	Female directors less overestimate merger gains and help create shareholder value

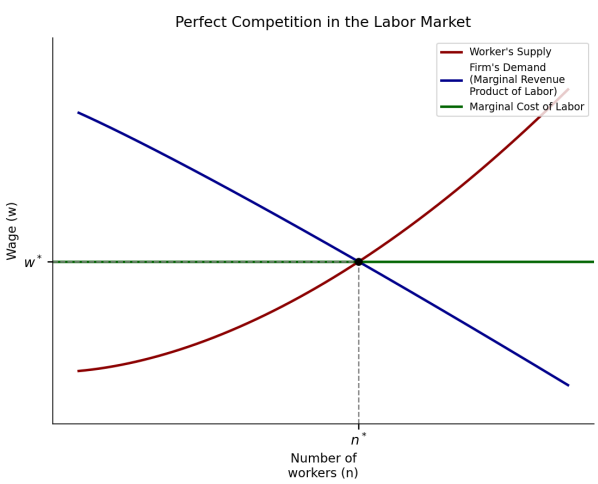
3 Theoretical Framework

To understand the gap between wages and workers' marginal productivity, a useful starting point is the benchmark case of a competitive labor market. As shown in Panel (a) of Figure 1, under perfect competition the marginal cost of labor is constant and equal to the market wage, which each firm takes as given. Firms therefore hire labor up to the point $n*$ where the marginal revenue product of labor (MRPL) equals the wage $w*$.

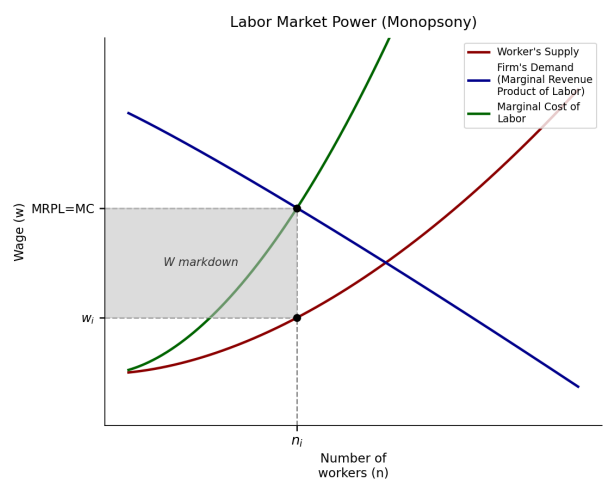
Monopsony power can be exercised by any employer who faces an upward-sloping labor supply curve, as shown in Panel (b) of Figure 1. Hiring an additional worker requires raising the wage not only for the marginal worker but, when wages are set uniformly, for all inframarginal workers as well. As a result, the marginal cost of labor lies strictly above the labor supply curve. The profit-maximizing firm sets employment where MRPL = MCL, yielding both lower employment and a lower wage relative to the competitive benchmark. Crucially, at the optimum, workers are paid a wage w_i determined by the labor supply curve, yet their marginal contribution to firm revenue satisfies MRPL $> w_i$. The wedge between the marginal revenue product of labor and the wage is known as the *markdown*, illustrated by the shaded region in Panel (b), and it is the central object of interest in this paper.

Figure 1: Labor Markets

(a) Perfect Competition



(b) Monopsony



To see how this markdown arises formally, following Ashenfelter et al. (2010) consider a firm's profit maximization problem with labor as the only input. Under perfect competition, the firm takes the market wage $\bar{w}$ as given and chooses labor input L to maximize profits:

$$\max_L \Pi = R(L) - \bar{w}L, \tag{1}$$

Where $R(L) \equiv P * Q(L)$ is the firm's revenue as a function of labor input. The first-order condition for profit maximization equates the marginal revenue with the marginal cost such that:

$$R'(L) = \bar{w} \tag{2}$$

Under monopsony, the firm faces an upward-sloping labor supply curve, so the wage is an increasing function of employment $w(L)$, as shown in figure 1. The profit maximization problem becomes:

$$\max_L \Pi = R(L) - w(L)L, \tag{3}$$

with first-order condition:

$$R'(L) = w(L) + w'(L)L \tag{4}$$

Note that the monopolistic firms also maximizes profits by equating marginal revenue and marginal cost, but the marginal cost of labor is greater than the wage in this case. The right-hand side of the equation exceeds the wage $w(L)$ by the term $w'(L)L$ which captures the additional cost of raising wages for all inframarginal workers when hiring one more worker.

Rearranging yields to the Lerner index formula for the labor market:

$$\frac{R'(L) - w(L)}{w(L)} = \frac{w'(L)L}{w(L)} \equiv \varepsilon_L^{-1} \tag{5}$$

Where $\varepsilon_L^{-1} \equiv \frac{w'(L)L}{w(L)}$ denotes the firm's perceived inverse elasticity of labor supply, capturing how much the wage must rise to attract an additional worker. This expression implies that the percentage gap between the marginal revenue product of labor and the wage equals the inverse of the perceived labor supply elasticity: the more inelastic the labor supply (i.e., the lower ε_L), the larger the gap.

From this, we define the main variable of interest, the labor markdown ν, following Yeh et al. (2022), as the ratio of the marginal revenue product of labor to the wage:

$$\nu \equiv \frac{R'(L)}{w(L)} = \varepsilon_L^{-1} + 1 \tag{6}$$

Under this definition, $\nu = 1$ under perfect competition (i.e., infinite labor supply elastic-

ity) and $\nu > 1$ when labor supply is upward-sloping. This measure, popularized by Manning (2013), captures the extent of monopsony power: a larger markdown ν indicates a greater gap between workers' marginal revenue product and their wage, and thus a higher degree of monopsony exploitation.

4 Data

This paper uses firm-level data from the Annual Manufacturing Survey (*Encuesta Anual Manufacturera*, EAM) conducted by Colombia's National Administrative Department of Statistics (DANE). The EAM is a census of medium and large manufacturing firms with more than 10 employees that provides detailed and standardized information on firms' production, input use, employment composition, and financial accounts. I restrict the analysis to the period 2000–2019 to ensure data consistency and to avoid disruptions associated with the COVID-19 pandemic and exclude the information before 2000 as the data in that period had limited information on executive positions by gender. Although the data are collected at the establishment level, I aggregate to the firm level by summing across all plants belonging to the same firm in a given year using the unique firm identifier provided in the survey, thereby capturing the full scope of each firm's operations as some managers might operate only from the headquarters office of the firm but still have impact on the plants.

Administrative firm-level data frequently contain extreme values arising from measurement error, recording mistakes, or genuine but highly influential outliers, which can bias production-function estimates and, by extension, the labor market power calculations that follow. I apply a series of sample restrictions to address this concern.

Following Yeh et al. (2022), I apply three filters to the key variables: output, labor, capital, and materials. First, I winsorize each variable at the top and bottom 1 percent of its distribution to limit the influence of extreme observations. Second, I drop observations with non-positive values for output or any input, as these are inconsistent with the production framework. Third, I compute each firm's materials cost share,defined as materials expenditure as a share of total costs, and exclude observations falling below one quarter of the sector-year median. Firms with anomalously low materials shares are unlikely to satisfy the conditions required for identification: no adjustment costs, static input choice, and the absence of input market power. Their inclusion could therefore contaminate the proxy-variable estimator.

Beyond these filters, I drop observations in which the total number of owners or directors is zero. Taken together,these restrictions exclude observations representing 7.99 percent of total gross output in the raw sample, making the final dataset still representative of the

manufacturing sector in Colombia. The final dataset is an unbalanced panel of 11,235 Colombian manufacturing firms observed over 2000–2019, yielding 53,688 firm-year observations spanning 105 industries and 23 geographic regions.

Table 2: Descriptive Statistics

Variable	Mean	SD	Min	Median	Max
Share of female executives (%)	45.192	39.925	0.000	49.812	100.000
Firm's share of industry employment (%)	0.020	0.066	0.000	0.004	1.000
Share of workers in production (%)	61.951	21.743	0.000	66.667	100.000
Gross output ($ millions USD)	1.219	536.010	756.959	11.356	2,968.214
Firm age (years)	4.262	3.422	1	3	20
Number of workers	56	146	1	20	4921
Wage per worker ($ thousands USD)	1.827	91	349.180	2.296	1,498.014

Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019. Note: The wage variable corresponds to the total wage bill of the firm divided by the number of employees, while gross output is proxied by total sales at factory price without direct taxes. To simplify the presentation, all dollar values are calculated using an exchange rate of 1 USD = 3,670 COP.

Table 2 reports descriptive statistics for the main variables used in the empirical analysis. The central variable of interest is the share of female executives, defined as the proportion of women among the firm's owners and senior managers. The sample mean stands at 45.2 percent, with a median of 49.8 percent, suggesting that the distribution is approximately symmetric. Nonetheless, the standard deviation of 39.9 percentage points indicates considerable cross-sectional heterogeneity. Firms in the sample are, on average, small. The mean (median) number of workers is 56 (20), with a standard deviation of 146 and a maximum of 4,921 employees, revealing a right-skewed size distribution consistent with the fact that manufacturing sectors in developing economies are dominated by small and medium-sized enterprises (Alviarez et al. (2025); Eslava et al. (2022)).

I measure each firm's labor market size using its share of industry employment, defined at the four-digit ISIC level. Formally, for firm i in industry j and year t:

$$\text{firm's share of industry employment}_{ijt} = \frac{L_{ijt}}{\sum_{k \in j} L_{kjt}} \tag{7}$$

where L_{ijt} denotes total employment of firm i in industry j in year t, and the denominator sums employment across all firms k operating in industry j in year t. The sample average of firm's share of industry employment is 0.02 percent, with a median of 0.004 percent, confirming that most firms are tiny in terms of employment relative to their respective industries. The large standard deviation (0.066 percentage points) and maximum value (1 percent) indicate, however, that a small number of firms command disproportionate shares of sectoral employment—a pattern consistent with the firm concentration patterns documented

by Alviarez et al. (2025) and Eslava et al. (2022). As markdowns are expected to be larger in more concentrated markets, this variable is crucial for the empirical analisys to isolate the effect of the labor market structure.

I also construct the share of workers engaged in production activities for each firm and year. The EAM distinguishes among four employment categories: production workers, professionals and technicians in non-production areas, and administrative, sales, and managerial staff. I define the production worker share as:

$$\text{Share of workers in production}_{it} = \frac{L_{it}^{prod}}{L_{it}^{prod} + L_{it}^{prof} + L_{it}^{adm}} \tag{8}$$

where L_{it}^{prod} denotes production workers, L_{it}^{prof} denotes professionals and technicians in non-production activities, and L_{it}^{adm} denotes administrative, sales, and managerial employees of firm i in year t. On average, 62 percent of the workforce is engaged in production activities, with a median of 66.7 percent and a standard deviation of 21.7 percentage points. The range spanning from 0 to 100 percent reflects meaningful variation in the degree to which firms are production-intensive versus administration-intensive, which may proxy for differences in technology, capital intensity, or organizational complexity across the sample.

Gross output is the total sales at factory prices net of direct taxes, and the average wage per worker is computed as total labor compensation divided by total employment. Both variables are reported in the EAM in Colombian pesos (COP) and are converted to U.S. dollars using a fixed exchange rate of 1 USD = 3,670 COP to facilitate interpretation.[2] Mean gross output is \$1.22 million USD, but the distribution is heavily right-skewed: the median of \$11.36 thousand USD lies far below the mean, and the standard deviation of \$536.01 million USD is orders of magnitude larger than the central tendency, pointing to the presence of a small number of very large firms that dominate the upper tail. This pattern is a well-established feature of firm-level data and underscores the importance of controlling for firm size in all empirical specifications as Yeh et al. (2022) found that this variable is significant for the estimation of markdowns. The average wage per worker is \$1.83 thousand USD annually, with a median of \$2.30 thousand USD. The dispersion in wages—ranging from \$0.35 to \$1,498.01 thousand USD—likely reflects both industry-level wage premia and substantial differences in skill composition and capital-labor ratios across firms.

The average firm in the sample is approximately 4.3 years old, with a median of 3 years and a maximum of 20 years. The right skew in the age distribution, combined with a standard deviation of 3.4 years, suggests that the sample is populated predominantly by young and recently established firms. This is consistent with high rates of entry and exit

[2]This exchange rate corresponds to the value in April 2nd 2026 and is applied uniformly across years.

that are characteristic of manufacturing sectors in emerging market economies (Alviarez et al., 2025), and motivates the inclusion of firm age as a control variable to account for lifecycle effects even though Yeh et al. (2022) found that it is not significant for the U.S. manufacturing sector.

Taken together, the descriptive statistics paint a picture of a sample dominated by small, young, production-intensive firms exhibiting wide variation in size, wages, output, and leadership composition. The pronounced dispersion across all dimensions underscores the importance of the empirical framework developed in the following sections, which is designed to exploit within-firm variation over time while controlling for the rich observed heterogeneity documented here.

5 Markdown Empirical Estimation

The theory section showed that the labor markdownν can be expressed as a function of the firm's perceived labor supply elasticity ε_L as $\nu = \varepsilon_L^{-1} + 1$. For the empirical estimation, Yeh et al. (2022) adopt the production approach of De Loecker et al. (2020), which allows for the estimation of firm-level markdowns using data on output and input costs without requiring direct information on market structure or the source of monopsony.

To illustrate this approach, consider a Cobb-Douglas production function with three inputs: labor L, capital K, and materials M, where L and M adjust frictionlessly within a period, while capital is subject to adjustment frictions. For simplicity, time and firm subscripts are omitted, though all variables are indexed by both dimensions. A key assumption of this framework is that materials (M) constitute a fully flexible input. Specifically, materials face no within-period adjustment costs, their price is taken as given by the firm, and their demand is chosen statically in the same period as production.

The firm's cost minimization problem is:

$$\mathcal{L} = rK + w(L)L + p^m M - \lambda \left(P\bar{Q} - AK^{\theta^K} L^{\theta^L} M^{\theta^M} \right) \tag{9}$$

where θ^K, θ^L, θ^M are the output elasticities of capital, labor, and materials, respectively and A denotes firm-specific productivity. The price of capital is r, the wage function is $w(L)$, and the price of materials is p^m. P is the price of output, $\bar{Q}$ is the target quantity of output.

The first-order conditions for M and L are:

$$\frac{\partial \mathcal{L}}{\partial M} = p^m - \lambda \theta^M A K^{\theta^K} L^{\theta^L} M^{\theta^M - 1} = 0. \tag{10}$$

$$\frac{\partial \mathcal{L}}{\partial L} = w'(L)L + w(L) - \lambda \theta^L A K^{\theta^K} L^{\theta^L - 1} M^{\theta^M} = 0, \tag{11}$$

We can simplify the first-order conditions as:

$$p^m = \lambda \theta^M \frac{Q}{M} \tag{12}$$

$$w'(L)L + w(L) = \lambda \theta^L \frac{Q}{L} \tag{13}$$

The Lagrange multiplier λ represents the marginal cost of production; therefore, the firm's markup,defined as the ratio of price to marginal cost, is given by:

$$\mu = \frac{P}{\lambda}. \tag{14}$$

Substituting and rearranging in the first-order conditions for materials:

$$p^m = \theta^M \frac{PQ}{M\mu}, \tag{15}$$

We obtain an expression for the markup in terms of observable quantities:

$$\mu = \theta^M \frac{PQ}{p^m M}, \tag{16}$$

Substituting $\mu = \frac{P}{\lambda}$ into the first-order condition for labor, and expressing the left-hand side in terms of the labor supply elasticity $w'(L)L + w(L) \equiv \frac{w'(L)L}{w(L)} + 1 \equiv (\varepsilon_L^{-1} + 1)w(L)$ we obtain:

$$(\varepsilon_L^{-1} + 1)w(L) = \frac{\theta^L}{\mu} \frac{PQ}{L}, \tag{17}$$

Now, we know from the theory section that the labor markdown ν is defined as the ratio of the marginal revenue product of labor to the wage, which can be expressed as:

$$\nu \equiv \frac{\text{MRPL}}{w(L)} = \varepsilon_L^{-1} + 1. \tag{18}$$

Substituting $\varepsilon_L^{-1} + 1$ with ν into equation (15) yields:

$$\nu = \frac{\theta^L}{\mu} \frac{PQ}{Lw(L)}. \tag{19}$$

Lastly, replacing the expression for the markup μ derived in equation (14) into the above equation for markdown ν we get to a expression for the markdown (productivity-wage gap)

which can be estimated using the ratio of the firm's expenditure on materials to its wage bill, scaled by the ratio of output elasticities:

$$\nu = \frac{\theta^L}{\theta^M} \cdot \frac{p^m M}{w(L)L}. \tag{20}$$

where $p^m M$ and $w(L)L$ can be extracted directly from the dataset, $p^m M$ is the total expenditure on raw materials, and $w(L)L$ is the total wage bill which corresponds to the total remuneration paid to permanent and temporary employees, both directly hired and outsourced. To estimate θ^L and θ^M, I use De Loecker et al. (2020) code which uses a translog production function with polynomial terms as follows:

$$\begin{aligned}\log(GO_{it}) = \beta_0 &+ \beta_1 \log(K_{it}) + \beta_2 \log(L_{it}) + \beta_3 \log(M_{it}) \\ &+ \beta_4 (\log K_{it})^2 + \beta_5 (\log L_{it})^2 + \beta_6 (\log M_{it})^2 \\ &+ \beta_7 \log(K_{it}) \log(L_{it}) + \beta_8 \log(K_{it}) \log(M_{it}) + \beta_9 \log(L_{it}) \log(M_{it}) \\ &+ \gamma_t + \varepsilon_{it}.\end{aligned} \tag{21}$$

where GO_{it} is gross output, K_{it} is capital, L_{it} is labor, M_{it} is intermediate inputs, γ_t are year fixed effects, and ε_{it} is the error term. Labor is measured as the total number of employees. Capital is proxied by the book value of fixed assets, with its rental cost computed via a user-cost formula that incorporates asset-specific depreciation rates from De Loecker et al. (2020) and year-specific interest rates and inflation from the Central Bank of Colombia (Banco de la República de Colombia, n.d.). Intermediate inputs are measured as total expenditure on raw materials, intermediate goods, and packaging; consistent with the requirements of the proxy-variable estimator, materials serve as the flexible input in production. Gross output is the firm-level total sales at factory prices, net of direct taxes.

The production function is estimated separately for each 4-digit ISIC sector via the two-stage GMM procedure of De Loecker et al. (2020), which corrects for the endogeneity of input choices.[3] Given the translog structure, output elasticities are firm-year specific and recovered as the partial derivatives of output with respect to each input:

[3]Because firms observe their own productivity before choosing inputs, OLS estimates are biased upward. The estimator recovers a productivity proxy $\hat{\phi}_{it}$ from a first-stage OLS regression, then constructs the productivity shock $\omega_{it}(\beta) = \hat{\phi}_{it} - \mathbf{x}'_{it}\beta$ for a candidate parameter vector β. Assuming a Markov process for productivity, its conditional expectation is approximated by a cubic polynomial in ω_{it-1}. The innovation $\xi_{it} = \omega_{it} - E[\omega_{it} \mid \omega_{it-1}]$ is orthogonal to lagged inputs by assumption, providing the GMM moment conditions. Capital is treated as predetermined and enters in current values; labor and materials are instrumented with their lagged values and lagged polynomial terms. An improvement applied to De Loecker et al. (2020) GMM criterion minimization code is the inclusion of a Nelder-Mead optimizer, with a gradient-based Rvmmin fallback when convergence is not achieved.

$$
\begin{aligned}
\hat{\theta}_{K,it} &= \hat{\beta}_1 + 2\hat{\beta}_4 \log(K_{it}) + \hat{\beta}_7 \log(L_{it}) + \hat{\beta}_8 \log(M_{it}), \\
\hat{\theta}_{L,it} &= \hat{\beta}_2 + 2\hat{\beta}_5 \log(L_{it}) + \hat{\beta}_7 \log(K_{it}) + \hat{\beta}_9 \log(M_{it}), \\
\hat{\theta}_{M,it} &= \hat{\beta}_3 + 2\hat{\beta}_6 \log(M_{it}) + \hat{\beta}_8 \log(K_{it}) + \hat{\beta}_9 \log(L_{it}).
\end{aligned} \tag{22}
$$

This paper implements a cascade approach: for sectors where the translog GMM fails to converge or yields non-positive elasticities, I fall back to a Cobb-Douglas specification estimated via the procedure of Ackerberg et al. (2015), which imposes sector-constant elasticities but is considerably more robust to limited panel variation.[4]

Table 3: Production Function Parameters

Variable	*Mean*	*p25*	*Median*	*p75*
$\hat{\theta}_{L,it}$ Output elasticity of labor	0.326	0.168	0.304	0.381
$\hat{\theta}_{K,it}$ Output elasticity of capital	0.117	0.066	0.101	0.143
$\hat{\theta}_{M,it}$ Output elasticity of materials	0.631	0.553	0.625	0.723
$\hat{\epsilon}_{it}$ Total factor productivity	0.573	2.333	3.285	3.968
$\hat{\mu}_{it}$ Markup	1.215	1.025	1.330	1.800
$\hat{\nu}_{it}$ **Markdown (productivity-wage gap)**	**0.835**	**0.576**	**1.680**	**3.242**

Source: Author's calculations based on the Colombian Manufacturing Survey (EAM) 2000–2019. Note: Output elasticities are estimated via the translog production function approach of De Loecker et al., 2020, with a Cobb-Douglas fallback following Ackerberg et al., 2015 for sectors where the translog GMM does not converge. Markups and markdowns follow Yeh et al., 2022.

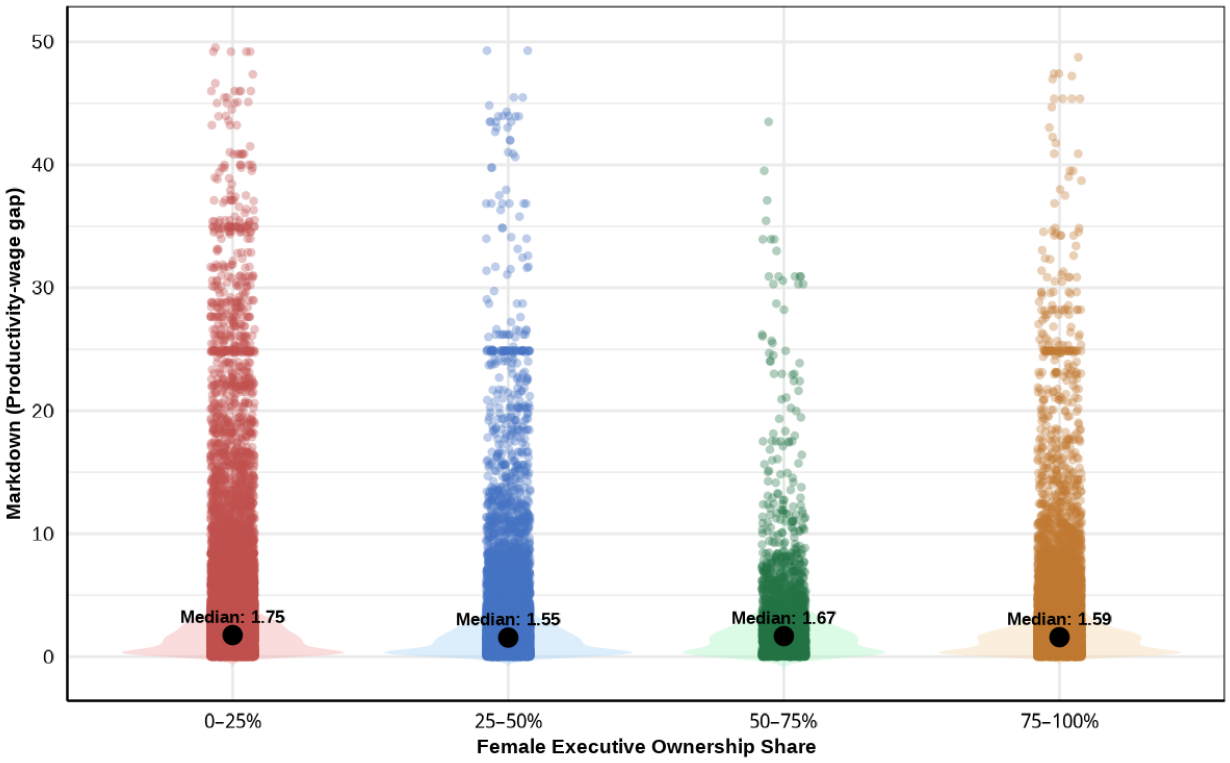
Table 3 reports the distribution of estimated output elasticities, markups, and markdowns. The estimated output elasticities are broadly consistent with the existing literature and lend credibility to the markdown estimates. The mean output elasticity of labor is 0.33 and the mean elasticity of materials is 0.63, with capital averaging 0.12, consistent with the capital-light, assembly-intensive nature of manufacturing in emerging markets. The dominance of materials in the cost structure further supports their use as the flexible input in the proxy-variable estimator. The mean product markup is 1.22, implying that the average firm prices output approximately 22 percent above marginal cost. The median of 1.33 falls within the range estimated by De Loecker et al. (2020) for U.S. manufacturing.

Turning to the markdown estimates, the median firm-level markdown is 1.68,implying that the median worker generates roughly 68 percent more revenue at the margin than they

[4]The Cobb-Douglas specification takes the form $\log(GO)_{it} = \beta_0 + \beta_K \log(K)_{it} + \beta_L \log(L)_{it} + \beta_M \log(M)_{it} + \gamma_t + \varepsilon_{it}$, estimated with the same two-stage GMM logic. Because elasticities are sector-constant under Cobb-Douglas, I impute the sector-level estimates to all firms in that sector. The vast majority of firm-year observations are covered by the translog estimator.

take home in wages — equivalently, for every dollar generated at the margin, the worker receives approximately 60 cents. This estimate is close to the findings of Alviarez et al. (2025), who report that workers in Colombia receive approximately 58 cents per dollar produced. This exceeds the median of 1.53 estimated for U.S. manufacturing by Yeh et al. (2022), consistent with stronger monopsony power in the Colombian labor market. The substantial dispersion in markdowns — the interquartile range runs from 0.58 to 3.24 — mirrors the firm-level heterogeneity documented in the literature (Yeh et al., 2022).

Figure 2: Markdown and Female Leadership



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM), 2000–2019.

Figure 2 presents how the distribution of markdowns varies by the share of female executives. A clear negative gradient emerges: firms where women hold between 0 and 25% of executive positions exhibit the highest median markdown (1.75), while firms with majority female executive representation (50-75% and 75-100%) display noticeably lower medians (1.67 and 1.59, respectively). The group with 25-50% female executive share sits at an intermediate level (1.55), reinforcing the downward trend. Taken together, these patterns suggest that more equitable female leadership is associated with workers being compensated closer to their marginal product of labor. This finding is consistent with the hypothesis that female executives may adopt compensation practices that reduce the degree of monopsonistic

wage-setting, narrowing the wedge between worker productivity and pay. While the distributions overlap considerably across groups reflecting the well-documented heterogeneity in firm-level markdowns, the systematic shift in medians provides suggestive evidence that the gender composition of firm leadership is meaningfully linked to labor markdowns.

6 Empirical Framework

To examine the relationship between labor market power—as measured by the firm-level productivity-wage gap— and female leadership, I estimate the following multiple linear regression model:

$$\log(\text{Markdown}_{it}) = \alpha + \beta\,\text{FemaleLeadership}_{it} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \alpha_t + \alpha_s + \alpha_r + \varepsilon_{it}, \tag{23}$$

The dependent variable $\log(\text{Markdown}_{it})$ is the logarithm of the productivity-wage gap for firm i in year t. The markdown is estimated using in the previous section using the methodology by Yeh et al. (2022) and is defined as the ratio of the firm's marginal revenue product of labor to its wage bill. The logarithmic transformation is applied to address positive skewness in the raw markdown distribution and obtain gaussian-like distributions in the error term ε_{it}.

The variable $\text{FemaleLeadership}_{it}$ denotes the share of women in executive and ownership positions at firm i in year t. The coefficient of interest, β, carries a semi-elasticity interpretation: it captures the approximate percentage change in the markdown associated with a one-percentage-point increase in the share of female executives, holding all other covariates constant. A negative estimate of β would indicate that greater female representation at the top of the firm hierarchy is associated with a compression of labor market power.

The vector $\mathbf{X}_{it}$ includes a rich set of firm-level controls designed to isolate the leadership channel from confounding variation in productivity, scale, and workforce composition. Firm size is captured through gross output deciles, which allow for a flexible, non-parametric relationship between scale and markdown. This addresses the findings from Yeh et al. (2022) that larger firms possess greater wage-setting capacity through both product-market scale effects and a higher share of firm-specific rents. Firms in the first decile serve as the reference category.

Firm age enters as a set of categorical indicators grouping firms into early (2-5 years), established (6-10 years), mature (11-15 years), and old (16-20 years) cohorts, with firms aged one year or less as the omitted category. This categorical specification is preferred over a linear age term because lifecycle dynamics in labor market power need not be monotonic:

older firms may accumulate monopsony rents through reputation, reduced worker mobility, and heightened attachment to firm-specific capital Total factor productivity (TFP), recovered from the production function estimation, is included to address the concern raised by Moreno-Gómez and Calleja-Blanco (2018) that female-led firms in Colombia are systematically more productive. Omitting TFP would conflate the female leadership effect with a productivity-level effect.

Labor market share, defined as the firm's share of industry-level employment, directly proxies for the firm's labor market power at the local-industry level and enters alongside the markdown to account for the possibility that the dependent variable partially reflects structural market thickness rather than leadership practices. Product market concentration is measured by the industry-level Herfindahl-Hirschman Index (HHI). The HHI is calculated as $HHI_j = \sum_{i=1}^{N_j} s_{ij}^2$ where s_{ij} denotes firm i's market share in industry j, and N_j is the total number of firms in that industry. This is used to control for the well-documented positive association between output-market concentration and wage markdowns (Berger et al., 2022). Finally, the share of production workers controls for heterogeneity in workforce composition across firms, which may independently affect both the estimated markdown — through the composition of the labor input in the production function — and the share of female executives.

The specification includes three sets of fixed effects. Year fixed effects (α_t) absorb aggregate macroeconomic fluctuations, including business-cycle variation in labor market tightness and economy-wide trends in productivity and wages. Sector fixed effects (α_s) account for permanent cross-industry differences in production technology, the degree of product market competition, and the structural propensity toward labor market concentration. Region fixed effects (α_r) control for persistent local labor market conditions, including geographic variation in workers' outside options, commuting costs, and the density of competing employers. In the most demanding specification, all three sets of fixed effects are included simultaneously, so that identification of β comes from within-sector, within-region, within-year variation in female leadership and markdowns across firms. Standard errors are clustered at the firm level throughout to account for serial correlation in the markdown within firms over time.

7 Results

Table 4: Regression Results - Dependent variable: log(Markdown)

	(1)	(2)	(3)	(4)
Share of Female Executives	-0.0010***	-0.0010***	-0.0011***	-0.0003 .
	(0.0001)	(0.0002)	(0.0002)	(0.0002)
Firm's share of employment in the industry		2.0356***	2.4268***	-1.0845***
		(0.3723)	(0.3713)	(0.3198)
Share of production workers		-0.0021***	-0.0018***	-0.0013***
		(0.0003)	(0.0003)	(0.0003)
Industry HHI		0.5729***	0.5719***	0.0913 .
		(0.0634)	(0.0635)	(0.0547)
Firm GO decile 2		0.0361	0.0510.	0.0104
		(0.0290)	(0.0288)	(0.0223)
Firm GO decile 3		0.0171	0.0328	-0.0422 .
		(0.0288)	(0.0286)	(0.0222)
Firm GO decile 4		-0.0428	-0.0159	-0.1071***
		(0.0291)	(0.0290)	(0.0225)
Firm GO decile 5		0.0733*	0.1043***	-0.0818***
		(0.0298)	(0.0296)	(0.0231)
Firm GO decile 6		0.1508***	0.1853***	-0.0239
		(0.0304)	(0.0303)	(0.0237)
Firm GO decile 7		0.1968***	0.2385***	0.0170
		(0.0313)	(0.0312)	(0.0245)
Firm GO decile 8		0.3773***	0.4155***	0.1575***
		(0.0325)	(0.0324)	(0.0256)
Firm GO decile 9		0.4822***	0.5162***	0.2371***
		(0.0336)	(0.0336)	(0.0266)
Firm GO decile 10		0.6640***	0.6782***	0.4416***
		(0.0391)	(0.0390)	(0.0315)
Age: Early (2–5 yrs)		-0.0383	-0.0554*	-0.0456*
		(0.0267)	(0.0265)	(0.0205)
Age: Established (6–10 yrs)		-0.0170	-0.0550.	-0.0143
		(0.0294)	(0.0293)	(0.0228)
Age: Mature (11–15 yrs)		0.0932*	0.0253	0.0471
		(0.0389)	(0.0389)	(0.0302)
Age: Old (16–20 yrs)		0.1025	0.0091	0.1161*
		(0.0683)	(0.0681)	(0.0528)
TFP		0.1557***	0.1578***	0.0801***
		(0.0039)	(0.0039)	(0.0036)
FE: Year	X	X	X	X
FE: Region	X		X	X
FE: Industry	X			X
N	48,292	31,664	31,664	31,664
R^2	0.5097	0.1243	0.1386	0.4849

Note: $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{\cdot}p < 0.1$

Table 4 reports the estimated coefficients from equation (23) across four progressively demanding specifications. Columns (1) through (3) sequentially add firm-level controls and region fixed effects while retaining year fixed effects throughout, Column (4) includes the full vector of firm-level controls alongside year, region, and industry fixed effects simultaneously.

The estimated coefficient on the share of female executives is negative and statistically significant at the 0.1% level in Columns (1) through (3), and remains negative — though attenuated — in the fully saturated specification in Column(4). In Column (3) a 10 percentage-point increase in the female executive share is associated with a 1.1 percent reduction in the log markdown ($\hat{\beta} = -0.0011$, $p < 0.001$).

To gauge the economic magnitude, consider the median firm in the sample, whose markdown of 1.680 implies that workers receive approximately 59.5 cents for every dollar of value they generate. A ten-percentage-point increase in female leadership compresses the markdown to approximately 1.662 —computed as $1.680x \exp(-0.0011 \times 10)$— raising the worker's share to roughly 60.2 cents per dollar produced, a gain of approximately 0.7 cents on the dollar, or \$7 for every \$100 of output, holding all other firm characteristics constant.

The coefficient attenuates to -0.0003 and loses significance at the 5% level once industry fixed effects are introduced in Column (4), though it remains marginally significant at the 10% level. The attenuation is informative: it indicates that a meaningful share of the cross-sectional association between female leadership and markdowns operates through the systematic sorting of female-led firms into industries that are structurally characterized by lower labor market power — for instance, more competitive, labor-intensive sectors. Once permanent industry-level heterogeneity in technology and market structure is absorbed, the residual within-industry association is smaller but directionally consistent. The stability of the sign across all four specifications, and the orderly attenuation pattern as identifying variation is progressively narrowed, is consistent with a genuine negative relationship rather than a statistical artifact of omitted industry confounders.

The gross output decile indicators reveal a robust, monotonically increasing relationship between firm size and the log markdown. Relative to firms in the bottom decile, firms in the top decile exhibit a log markdown that is 0.44 log points higher in the preferred specification, a differential that is highly significant and economically large. The relationship becomes statistically distinguishable from zero beginning in the eighth decile and accelerates sharply in the upper tail of the size distribution. This pattern is consistent with the theoretical prediction that larger employers face less elastic labor supply curves, as workers bear higher search and relocation costs when fewer alternative employers are available locally, thereby affording large firms greater wage-setting power (Card, 2022; Manning, 2021; Yeh et al.,

2022).

The TFP coefficient is positive and highly significant across all specifications, confirming that more productive firms pay workers a smaller fraction of their marginal product. This is consistent with a rent-extraction mechanism in which productivity gains accrue disproportionately to firm owners rather than being passed through to wages, as documented in the broader literature on firm-level rent sharing (Card, 2022). The coefficient attenuates by roughly half — from 0.158 in Column (3) to 0.080 in Column (4) — once industry fixed effects are included, indicating that a substantial portion of the productivity-markdown gradient reflects persistent cross-industry differences in production technology rather than purely within-industry variation.

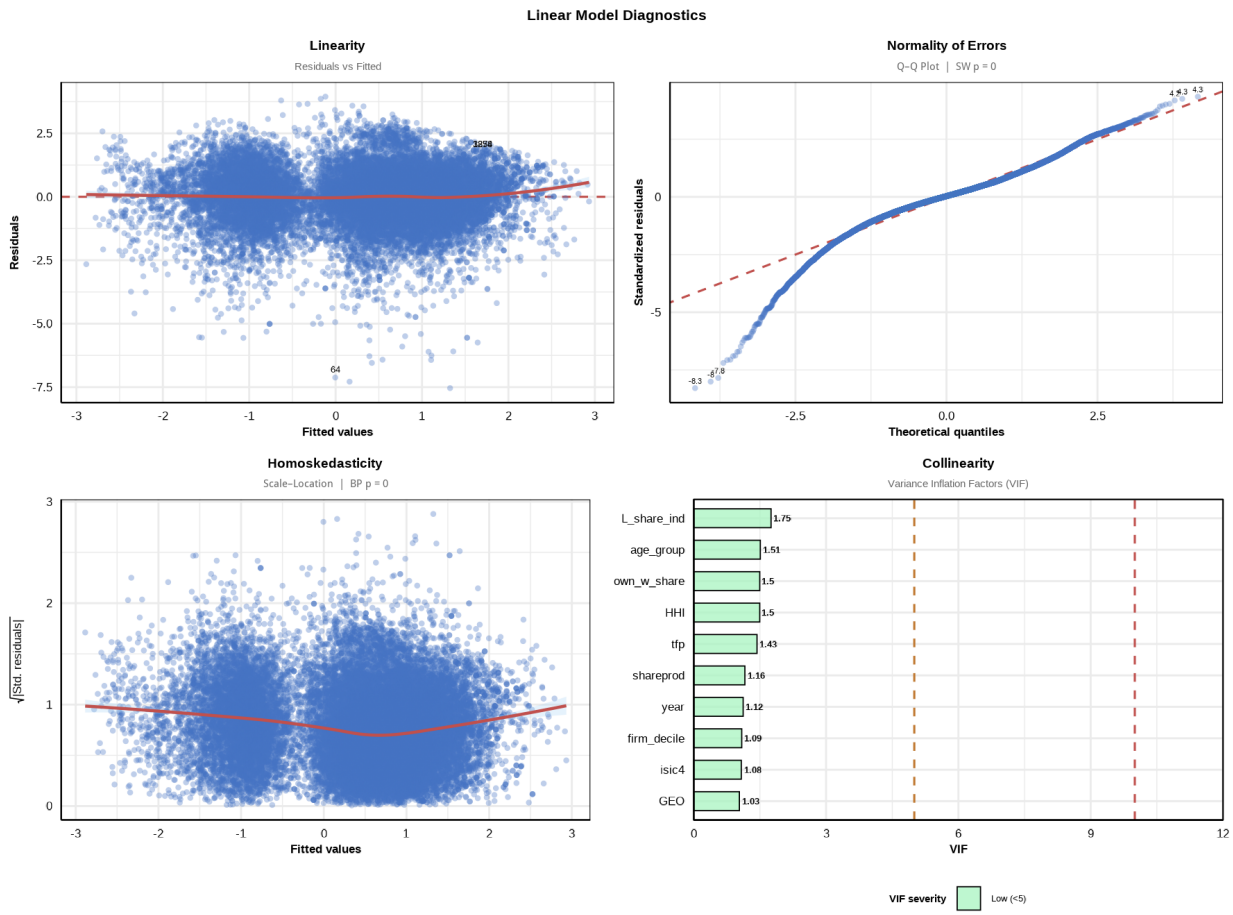
The industry HHI enters positively and significantly in Columns (2) and (3), consistent with the view that product market concentration amplifies labor market power by reducing the effective number of competing employers and facilitating tacit or explicit wage coordination (Azar et al., 2017; Berger et al., 2022). The coefficient contracts to marginal significance in Column (4) once industry fixed effects absorb the time-invariant component of market concentration. This is the expected pattern: the HHI is largely a time-stable, industry-level variable, and its identifying variation is substantially exhausted by the industry fixed effects. The residual marginal significance suggests a modest within-industry, time-varying component of concentration that continues to predict markdown variation.

Relative to the youngest cohort (firms aged one year), early-stage firms (2-5 years) exhibit lower markdowns in the preferred specification ($\hat{\gamma} = -0.046$, $p < 0.05$), consistent with the notion that very young firms may initially underprice labor to attract workers before accumulating the reputation and worker attachment that sustain monopsony rents over time cite someone. Coefficients for older age cohorts are positive but imprecisely estimated in most specifications, suggesting diminishing lifecycle effects beyond the early growth phase. The non-monotonic pattern across intermediate age cohorts reinforces the decision to model age as a categorical variable rather than imposing a linear lifecycle gradient.

The R^2 rises from 0.139 in Column (3) to 0.485 in the fully saturated Column (4). It bears noting that R^2 is non-decreasing in the number of regressors by construction, so mechanical inflation from the addition of industry fixed effects cannot be ruled out as a partial contributor to this increase. That said, the large increment underscores that permanent industry-level heterogeneity — in production technology, the degree of product market competition, and the structural propensity for labor market concentration — is the dominant source of cross-firm variation in labor market power, and motivates the use of the fully saturated specification as the preferred benchmark for inference on the female leadership coefficient.

7.1 Gauss-Markov Multiple Regression Assumptions

Figure 3: Model Diagnostics - Column (4)



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM), 2000–2019.

Figure 3 presents the four standard diagnostic plots for Column (4), evaluating the classical assumptions underlying the OLS estimator. The residuals-versus-fitted plot (top left) assesses the assumption that the conditional expectation of the error is zero. The fitted LOESS smoother remains virtually indistinguishable from the zero line across the full support of fitted values, providing no evidence of systematic nonlinearity or misspecified functional form. The scale-location plot (bottom left) evaluates homoskedasticity by mapping the square root of standardized residuals against fitted values. Although the residual spread appears relatively stable visually, the Breusch–Pagan test rejects the null of constant error variance ($p = 0$), confirming the presence of heteroskedasticity. This does not invalidate the OLS point estimates — which remain unbiased and consistent under heteroskedasticity — but it does invalidate conventional standard errors. This concern is addressed throughout by

clustering standard errors at the firm level, which yields variance estimates that are robust to both arbitrary heteroskedasticity and within-firm serial correlation. The Q–Q plot (top right) examines normality of errors by comparing standardized residuals against theoretical normal quantiles. The bulk of the distribution tracks the 45-degree reference line closely, confirming approximate normality in the central mass; visible departures in the lower tail reflect a small number of firms with extreme productivity-wage gaps. At the sample size of $N = 31{,}664$, the central limit theorem ensures that the sampling distribution of $\hat{\beta}$ is approximately normal regardless of the tail behavior of ε_{it}, so this departure does not threaten inference. Multicollinearity is assessed via Variance Inflation Factors (bottom right panel). All regressors exhibit VIF values below 2 — the largest being 1.75 for the firm's labor market share — well below the conventional warning thresholds of 5 and 10, indicating that collinearity among covariates neither inflates standard errors nor destabilizes coefficient estimates. Finally, the assumption of a random sample is satisfied by construction: as discussed in Section 4, the Colombian Annual Manufacturing Survey covers the universe of manufacturing establishments above the statutory size threshold, and the analytic sample is drawn from this population using a consistent sampling frame across the 2000–2019 panel. Taken together, the diagnostics support the validity of the OLS estimator for the preferred specification: the point estimates are unbiased and consistent, the clustered standard errors are heteroskedasticity-robust, and inference is asymptotically valid.

8 Conclusion

This paper examines whether the gender composition of firm leadership shapes the extent to which Colombian manufacturing firms exploit labor market power. Using firm-level panel data from the Annual Manufacturing Survey (2000–2019) and firm-specific productivity-wage gaps estimated via the production-function methodology of Yeh et al. (2022), I find a robust negative association between the share of female executives and the labor markdown. A ten-percentage-point increase in female leadership is associated with a 1.1 percent compression of the productivity-wage gap, a result that is stable across specifications controlling for firm size, TFP, workforce composition, and market concentration, and directionally consistent — though attenuated — once permanent industry-level heterogeneity is absorbed. Taken together, the findings suggest that management structure is a meaningful margin of variation in firms' wage-setting behavior, and that greater female representation at the top of the firm hierarchy is associated with less aggressive exploitation of monopsony rents — consistent with evidence that female leaders place greater weight on worker welfare and interpret productivity signals differently than their male counterparts (Flabbi et al., 2017; Matsa & Miller, 2013).

These findings carry implications for both policy and future research. On the policy side, they suggest that interventions promoting gender diversity in executive and ownership positions — including board gender quotas, bias-free promotion practices, mentorship pipelines, and pay equity audits — may generate positive labor market externalities beyond the firms directly affected, by moderating the translation of productivity into wages across the manufacturing sector. In the Colombian context, where labor market frictions are pronounced and the productivity-wage gap is substantial (Amodio & de Roux, 2024), such policies could meaningfully raise workers' share of output without requiring direct wage regulation. On the research side, the attenuation of the female leadership coefficient upon the inclusion of industry fixed effects points to an important open question: to what extent does the effect operate through the selection of female-led firms into more competitive industries versus through within-firm changes in wage-setting behavior? Addressing this question will require either quasi-experimental variation in leadership composition — such as that generated by board gender quota legislation — or matched employer-employee data that permit identification of gender-specific wage markdown heterogeneity within firms, two promising avenues for future work.

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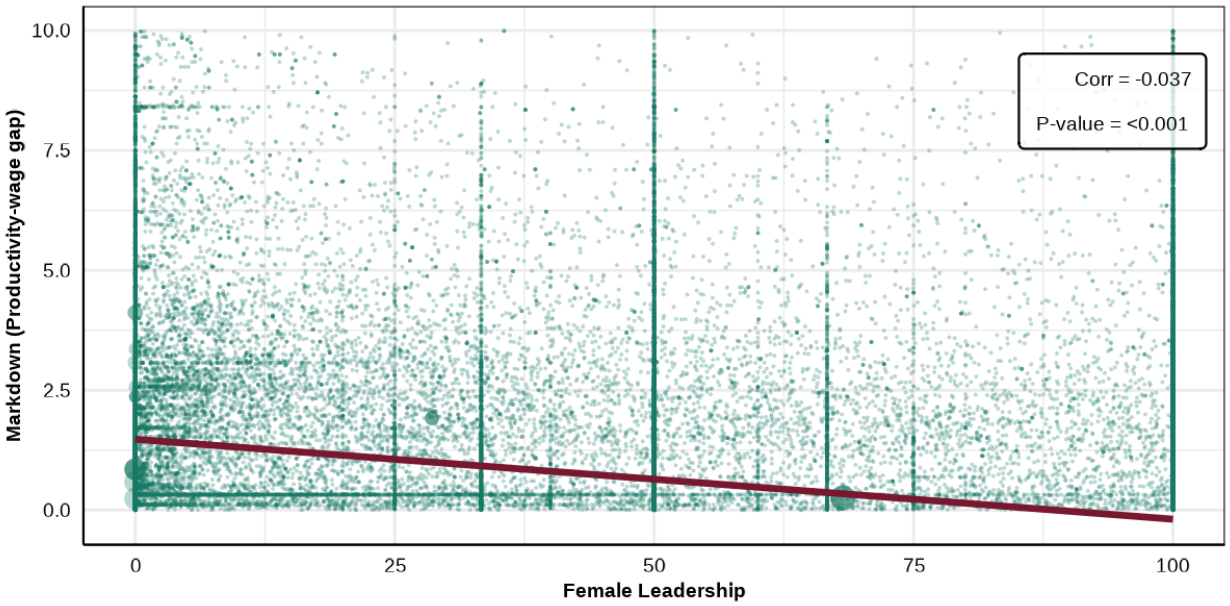
Appendix: Correlation between Markdown and Female Leadership

Figure 4 plots the firm-level estimated markdown against the female share of ownership, with scatter points weighted by each firm's gross output. The correlation is negative ($\rho = -0.037$, $p < 0.001$), offering preliminary evidence that greater female participation in leadership positions is associated with workers capturing a larger share of their marginal revenue product. For visual clarity, the figure is truncated at markdown values below 10; however, the reported correlation is computed over the full sample of observations.

Figure 4: Markdown and Female Leadership



Source: Author's calculations based on the Colombian Manufacturing Survey (EAM), 2000–2019.